

Weakly-Supervised 3D Anomaly Segmentation via Soft Mask-Coupled Diffusion Inpainting

Anonymous submission

Abstract

We present a weakly supervised framework for 3D medical anomaly segmentation that requires no voxel-level annotations at any stage, relying only on bounding boxes or centroids at inference time. A 3D denoising diffusion probabilistic model (DDPM) trained exclusively on healthy tissue patches inpaints a weak-label- initialized soft-mask region as plausible healthy content; the residual between the original and inpainted volumes localizes the anomaly. We introduce (i) a *soft mask-coupled reverse diffusion process* in which the anomaly mask evolves as a latent variable jointly with the image during sampling—the first such formulation in reconstruction-based anomaly detection—together with a multi-timestep discrepancy score and mask-scoped classifier guidance; and (ii) a *synthetic anomaly augmentation suite* paired with an *anomaly-aware hinge regularizer* that sharpen the healthy-tissue prior and improve cross-domain robustness. Validated on lung CT (LIDC-IDRI, NLST, LUNA16) with cross-organ (liver CT) and cross-modality (brain MRI) generalization experiments, our framework outperforms state-of-the-art interactive foundation models (nnInteractive, SegVol, SAM-Med3D, MedSAM2) specifically in the sub-6 mm, low-contrast regime where boundary-detection methods fail.

1 Introduction

Volumetric anomaly segmentation in 3D medical imaging is a prerequisite for radiomics analysis, malignancy stratification, and longitudinal monitoring. Fully supervised 3D segmentation models achieve strong performance on LIDC-IDRI, the only large-scale lung CT dataset with voxel-level mask annotations ($\approx 1,000$ cases). The vast majority of clinical 3D imaging datasets—including the National Lung Screening Trial (NLST, 26,254 subjects) and analogous repositories across liver, kidney, and brain—carry only cheap *weak labels*: bounding boxes, centroids, or RECIST diameters. This annotation bottleneck is universal across organ types and imaging modalities.

State-of-the-art interactive foundation models (Isensee et al. 2025; Du et al. 2024; Wang et al. 2023; Ma et al. 2025) accept box/point prompts but segment “the thing in the box”

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by boundary contrast: they have no mechanism to distinguish pathological from normal tissue. This fails critically for sub-6 mm nodules and ground-glass opacities (GGOs), where intensity boundaries are absent or ambiguous across all modalities.

We reframe weakly supervised anomaly segmentation as *healthy-tissue counterfactual generation* via 3D diffusion inpainting. Our model is trained only on healthy patches; at inference, given a weak label it inpaints the indicated region as plausible healthy tissue, and the residual localizes the anomaly. A novel *soft mask-coupled reverse process* refines this residual into a tight 3D pseudo-segmentation mask entirely inside the generative sampling dynamics. Because the model learns only “normal,” it is anomaly-agnostic and transfers across organs and modalities without retraining on lesion data.

Contributions. Our contributions span three groups:

Diffusion & Sampling (C1–C4). Soft Gaussian mask initialization from weak labels (C1) and a soft mask-coupled reverse diffusion kernel (C2), in which the anomaly mask m_t evolves as a latent variable inside the reverse process—the first such formulation in reconstruction-based anomaly detection. A multi-timestep discrepancy score (C3) exploiting the lesion-size-dependent information content of the per-step residual r_t , and mask-scoped classifier guidance (C4) that actively suppresses pathological reconstructions within the evolving mask.

Training (C5–C6). A four-component synthetic anomaly augmentation suite (C5) and an anomaly-aware hinge regularizer (C6) that sharpen the healthy-tissue prior and improve robustness to domain shift, theoretically coupled to C2 and C4 through the soft mask.

Generalisation (C7–C9). Cross-dataset generalisation on LIDC-IDRI, NLST, LUNA16 (C7); cross-organ transfer from lung to liver/kidney CT (C8); and cross-modality transfer from CT to brain MRI (C9)—demonstrating the framework is not lung- or CT-specific.

2 Related Work

Supervised 3D Lesion Segmentation. nnU-Net (Isensee et al. 2021) is the de facto upper bound for supervised CT segmentation, auto-configuring a 3D U-Net to any dataset.

Transformer-based architectures (SwinUNETR, nnFormer) improve on LIDC-IDRI but require full mask supervision and degrade sharply outside their training distribution, especially on sub-6 mm nodules due to extreme class imbalance.

Weakly Supervised Lesion Segmentation. DeepEM (Zhu et al. 2018) trains 3D CNNs from location labels via EM over latent boxes. RECIST-based supervision (Zhou et al. 2023) converts 2D diameters into 3D pseudo-labels. Most relevantly, Anonymous (2026b) use a 3D latent diffusion model’s generative guidance to suppress nodule signal in reconstruction on LUNA16—the closest existing work. Key differences: their model is not trained on healthy-only data; their mask is not evolved inside the reverse process; they provide no classifier guidance, regularizer, or cross-organ/modality evaluation.

Reconstruction-Based Anomaly Detection. AnoDDPM (Wyatt et al. 2022) introduced partial diffusion with simplex noise for brain MRI, training on healthy data and computing reconstruction error as the anomaly score (+25.5% Dice over f-AnoGAN). Extensions include latent-space partial diffusion (Graham et al. 2023), Synomaly noise for ultrasound (Anand and Tao 2024), counterfactual generation for brain tumours (Fontanella et al. 2024), IterMask2 (Liang et al. 2024), and FDP (Anonymous 2026a) for brain MRI. All are 2D or 2D-stacked, use a fixed corruption region, and do *not* evolve the mask inside the reverse process.

Adaptive Mask Inpainting. AMI-Net (Luo et al. 2024) proposes an adaptive mask generator for industrial anomaly detection (MVTec AD, BTAD). MAD-paint (Liu et al. 2025) introduces mask-uncertainty metrics for diffusion inpainting. Neither is 3D, medical, nor couples the mask inside the reverse diffusion kernel.

Foundation Models for Medical Segmentation. MedSAM (Ma et al. 2024) fine-tunes SAM on medical data but remains 2D. SAM-Med3D (Wang et al. 2023) re-trains SAM natively in 3D. SegVol (Du et al. 2024) is a universal interactive CT segmenter. MedSAM2 (Ma et al. 2025) applies SAM2’s video memory to 3D volumes. nnInteractive (Isensee et al. 2025) won the CVPR 2025 Foundation Models for Interactive 3D Biomedical Image Segmentation Challenge. All are boundary-detection paradigms lacking an anomaly-vs-normal distinction mechanism; all degrade on small/low-contrast lesions.

3 Method

Let $x_0 \in \mathbb{R}^V$ be a 3D CT patch (64^3 voxels) and (c, b) the weak label—centroid c and optional box b . Standard DDPM (Ho, Jain, and Abbeel 2020) defines a forward process $q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$ and trains $\epsilon_\theta(x_t, t)$ to predict noise via $\mathcal{L}_{\text{simple}} = \mathbb{E}[\|\epsilon - \epsilon_\theta(x_t, t)\|^2]$. Our framework operates in three phases: (I) soft mask initialization; (II) soft mask-coupled reverse diffusion; (III) final pseudo-mask extraction.

I. Diffusion and Sampling Process

C1: Soft Mask Initialization (Fixed-Mask Baseline). Rather than inpainting a hard binary box, we initialize the

inpainting mask as a smooth Gaussian field centered on c , encoding spatial uncertainty about the true lesion boundary:

$$m_T(v) = \exp\left(-\frac{d(v, c)^2}{2\sigma^2}\right), \quad (1)$$

where σ is estimated from the box diagonal when b is available, or set to a dataset-prior default for centroid-only labels. This avoids hard box-edge artifacts in the residual map and provides a continuous field that C2’s coupled update can smoothly contract. The fixed soft-mask variant (no coupling) serves as the primary ablation baseline.

C2: Soft Mask-Coupled Reverse Diffusion. We extend the DDPM reverse transition to treat $m_t \in [0, 1]^V$ as a latent variable evolving jointly with x_t at every reverse step:

$$p_\theta(x_{t-1} | x_t) \longrightarrow p_\theta(x_{t-1}, m_{t-1} | x_t, m_t). \quad (2)$$

At each step $t \rightarrow t-1$, the per-voxel discrepancy between the noisy image and the model’s denoised estimate $\hat{x}_0(x_t, t)$ drives the mask update:

$$r_t(v) = |x_t(v) - \hat{x}_0(x_t, t)(v)|, \quad (3)$$

$$m_{t-1}(v) = \text{clip}\left(m_t(v) \cdot (1 - \lambda [r_t(v) < \tau_t]), 0, 1\right), \quad (4)$$

where λ is a decay rate and τ_t anneals with t . Voxels where the healthy-tissue model agrees with the observation (low r_t —reconstructible as healthy) have their mask weight decayed; voxels where it persistently disagrees (high r_t —not explainable by a healthy prior) retain high mask weight. The combined image update follows RePaint-style conditioning through the evolving mask:

$$x_{t-1}(v) = m_{t-1}(v) x_{t-1}^{\text{known}}(v) + (1 - m_{t-1}(v)) x_{t-1}^{\text{rev}}(v). \quad (5)$$

The final mask m_0 is the pseudo-segmentation directly—no external thresholding or post-processing loop is required. This is the first formulation in which the inpainting mask in reconstruction-based anomaly detection is a latent variable with its own update rule *inside* the reverse diffusion kernel.

C3: Multi-Timestep Discrepancy Score. r_t carries anomaly information across the full noise-level spectrum. Large t captures coarse structural anomalies (dominant for large lesions); small t captures fine boundary detail (critical for sub-6 mm nodules and GGOs). Drawing on the connection between per-step score norms and out-of-distribution likelihood (Song et al. 2021), we treat r_t as a multi-scale anomaly signal and compute the final anomaly score as a learned weighted aggregation of r_t across selected timesteps. We empirically validate this with an AUC-vs-timestep analysis stratified by lesion diameter, showing that the optimal timestep range shifts systematically with lesion size—explaining why AnoDDPM’s fixed- λ partial diffusion is sub-optimal for small lesions.

C4: Mask-Scoped Classifier Guidance. A patch-level binary classifier $p_\phi(y | x_t, t)$ trained on LIDC-IDRI distinguishes healthy from anomalous patches at all noise levels. During sampling, its gradient is injected into the reverse update, spatially scoped to the evolving soft mask m_t :

$$\hat{\epsilon}(x_t, t)(v) = \epsilon_\theta(x_t, t)(v) - s \cdot m_t(v) \cdot \sqrt{1 - \bar{\alpha}_t} \nabla_{x_t} \log p_\phi(y = \text{healthy} | x_t) \quad (6)$$

where s is the guidance scale. This compensates for residual anomaly leakage from the conditioning context that passive healthy-only training cannot suppress. The spatial scoping by m_t ensures guidance is zero in confirmed-healthy surroundings and maximal where mask weight is highest. The classifier gradient magnitude also provides a second independent anomaly signal fused with r_t .

II. Training: Augmentation and Regularization

C5: Synthetic Anomaly Augmentation Suite. We expose the model to four types of synthetic anomaly during training as structured data augmentation:

1. *Gaussian intensity blobs* ($\sigma \in [2, 15]$ mm, $\Delta\text{HU} \in [20, 400]$) simulating solid nodule HU signatures.
2. *Frequency-domain texture perturbations* simulating GGO low-contrast appearance.
3. *Perlin-deformed sphere primitives* producing lobulated and spiculated morphologies.
4. *Cross-organ cut-paste augmentation*: liver/kidney lesion crops intensity-normalized and pasted into healthy lung patches, providing cross-domain diversity.

Each component is ablated independently. These synthetic anomalies are used *only* as training augmentation—never as test-time annotations.

C6: Anomaly-Aware Hinge Regularizer. We add a hinge term that penalizes the model for reconstructing injected anomalies too accurately, sharpening the healthy-tissue prior:

$$\mathcal{L} = \mathcal{L}_{\text{simple}} + \gamma \mathbb{E}_{\tilde{x}_0} \left[\max \left(0, \delta - \|(\tilde{x}_0 - \hat{x}_0(\tilde{x}_t, t)) \odot m_{\text{syn}}\|^2 \right) \right], \quad (7)$$

where $\tilde{x}_0$ is a healthy patch with a synthetic anomaly in m_{syn} , γ controls regularization strength, and δ is the minimum required residual in anomaly regions. C5 and C6 are coupled: augmentation diversity determines the variety of injected anomalies; the regularizer determines the loss response. Together they sharpen the prior that C2’s discrepancy signal r_t and C4’s classifier gradient depend on.

III. Generalisation

Because the model trains only on healthy tissue and makes no assumptions about anomaly appearance, it is *anomaly-agnostic by construction*. **C7** (cross-dataset) is demonstrated on NLST and LUNA16/25 without retraining. **C8** (cross-organ) is demonstrated by training a healthy liver parenchyma prior and applying the same pipeline to MSD Task 03 liver tumours with box-only weak labels—providing a second full-mask evaluation dataset independent of lung CT. **C9** (cross-modality) trains a healthy brain MRI prior and evaluates on BraTS with box-only labels, compared directly against AnoDDPM, IterMask2, and FDP-AAAI2026 (Anonymous 2026a). For C8 and C9 only the healthy-tissue diffusion model is retrained on the new domain; all other components (soft mask coupling, regularizer, classifier guidance) are identical.

Table 1: Main segmentation results on LIDC-IDRI (Dice/IoU, full mask GT) and LUNA16 (FROC). Best weakly supervised result in **bold**; best overall underlined.

Method	Supervision	LIDC Dice	LUNA FROC
nnU-Net 3D	Full mask	[TODO:]	[TODO:]
MedSegDiff-V2	Full mask	[TODO:]	[TODO:]
AnoDDPM-3D	Unsupervised	[TODO:]	[TODO:]
VAE-3D AD	Unsupervised	[TODO:]	[TODO:]
MedSAM (2D)	Box prompt	[TODO:]	[TODO:]
SAM-Med3D	Box prompt	[TODO:]	[TODO:]
SegVol	Box prompt	[TODO:]	[TODO:]
MedSAM2	Box prompt	[TODO:]	[TODO:]
nnInteractive	Box prompt	[TODO:]	[TODO:]
Ours (Full)	Box/centroid	[TODO:]	[TODO:]

4 Experiments

Datasets

Lung CT. LIDC-IDRI (Armato et al. 2011) (1,018 scans, four-radiologist consensus voxel masks; *oracle evaluation*). NLST (The National Lung Screening Trial Research Team 2011) (26,254 subjects, box/centroid weak labels; *scale test*). LUNA16 (Setio et al. 2017) (*FROC benchmark*).

Cross-organ CT (C8). MSD Task 03 liver tumours (Simpson et al. 2019) or KiTS kidney tumours (Heller et al. 2023) (full mask GT; box-only weak labels at inference).

Cross-modality MRI (C9). BraTS (Menze et al. 2015) brain tumours (full mask GT; box-only weak labels at inference).

Baselines

We compare against: (i) nnU-Net 3D (Isensee et al. 2021) (supervised upper bound); (ii) MedSegDiff-V2 (Wu et al. 2024) (supervised diffusion segmentation, AAAI 2024); (iii) AnoDDPM reimplemented natively in 3D (Wyatt et al. 2022) (unsupervised reconstruction, no weak labels); (iv) VAE-based 3D anomaly detection (non-diffusion reconstruction baseline); (v) AMI-Net adapted to 3D CT (Luo et al. 2024); (vi) MedSAM (Ma et al. 2024) (2D, box-prompted, slice-by-slice); (vii) SAM-Med3D (Wang et al. 2023); (viii) SegVol (Du et al. 2024); (ix) MedSAM2 (Ma et al. 2025); (x) nnInteractive (Isensee et al. 2025) (all 3D, box-prompted; nnInteractive is the CVPR 2025 challenge winner).

Main Results

[TODO: Table 1: Dice/IoU on LIDC-IDRI and FROC on LUNA16 across all baselines. Fill after experiments complete.]

Stratified Analysis. Table 2 reports Dice stratified by nodule diameter, demonstrating our advantage in the sub-6 mm regime against all baselines including nnInteractive—the primary empirical claim of this paper.

Simulated Weak-Label Experiment

To directly validate our method in the annotation regime it targets, we strip LIDC-IDRI masks to bounding boxes/centroids

Table 2: Dice stratified by nodule diameter on LIDC-IDRI.

Method	< 6 mm	6–10 mm	> 10 mm
nnU-Net 3D	[TODO:]	[TODO:]	[TODO:]
nnInteractive	[TODO:]	[TODO:]	[TODO:]
SegVol	[TODO:]	[TODO:]	[TODO:]
Ours	[TODO:]	[TODO:]	[TODO:]

Table 3: Ablation on LIDC-IDRI Dice (LIDC) and sub-6 mm Dice (Small).

Configuration	LIDC Dice	Small
(0) Hard-box inpainting (baseline)	[TODO:]	[TODO:]
(1) + Soft Gaussian init (C1)	[TODO:]	[TODO:]
(2) + Mask-coupled reverse (C2)	[TODO:]	[TODO:]
(3) + Multi-timestep score (C3)	[TODO:]	[TODO:]
(4) + Classifier guidance (C4)	[TODO:]	[TODO:]
(5) + Synthetic augmentation (C5)	[TODO:]	[TODO:]
(6) + Hinge regularizer (C6) = Full model	[TODO:]	[TODO:]

and run the full pipeline using *only* these degraded labels, then evaluate against the real withheld masks. This addresses the primary reviewer concern: “does the method work when you actually only have weak labels?”

Ablation Study

Table 3 ablates each contribution in sequence, isolating the effect of each component.

Additional ablations cover: initial σ sensitivity; λ/τ_t annealing schedule; guidance scale s ; DDIM step count; and each augmentation component independently.

Cross-Organ and Cross-Modality Generalisation

Table 4 reports MSD liver Dice (C8) and BraTS Dice (C9) compared against AnoDDPM-3D, IterMask2, and FDP-AAAI2026 in the box-prompted weak-label regime. We also report downstream malignancy classification AUC from radiomic features extracted from NLST pseudo-masks, compared against features from LIDC-IDRI expert masks, directly validating the “weakly supervised segmentation for downstream radiomics” framing.

5 Scope, Limitations, and Conclusion

Scope

Within lung CT, the framework is anomaly-agnostic: solid nodules, part-solid nodules, GGOs, and any localized deviation from healthy parenchyma are detected via the same residual mechanism. Cross-organ and cross-modality scope is empirically demonstrated (C8, C9). Extension to ultrasound and PET requires domain-specific healthy-prior training and is reserved for future work. The framework is applicable to any clinical dataset with bounding box or centroid annotations and a healthy-tissue corpus for training.

Table 4: Generalisation across organs (liver CT) and modalities (brain MRI), box-prompted weak-label setting.

Method	Liver Dice (C8)	BraTS Dice (C9)	NLST AUC
AnoDDPM-3D	[TODO:]	[TODO:]	—
IterMask2	—	[TODO:]	—
FDP	—	[TODO:]	—
nnInteractive	[TODO:]	[TODO:]	—
Ours	[TODO:]	[TODO:]	[TODO:]

Limitations

Inference speed. Iterative reverse diffusion with per-step mask updates and classifier guidance is slower than a single network forward pass. DDIM acceleration is applied but per-nodule latency exceeds nnU-Net/nnInteractive. Consistency-model distillation is planned future work.

Classifier dependency. C4’s classifier is trained on LIDC-IDRI, introducing a semi-supervised element; it must be retrained per new organ/modality for C8/C9.

Patch size. The 64^3 patch-level framework may fragment very large or diffuse lesions at boundaries; multi-scale hierarchical inference is future work.

Modality scope. Cross-modality validation covers brain MRI only; ultrasound and PET are not evaluated in this paper.

Conclusion

We presented a unified framework for weakly supervised 3D anomaly segmentation requiring no voxel-level annotations at any stage. Our soft mask-coupled reverse diffusion (C1, C2), multi-timestep discrepancy score and classifier guidance (C3, C4), synthetic augmentation and hinge regularizer (C5, C6), and cross-dataset/-organ/-modality generalisation (C7–C9) constitute a coherent system coupled through the evolving soft mask m_t . Because the model is trained only on healthy tissue and never on any specific pathology, it is anomaly-agnostic by construction: the same pipeline detects solid nodules, GGOs, liver lesions, and brain tumours using only weak labels—without retraining on lesion data. The framework outperforms interactive foundation models specifically in the sub-6 mm, low-contrast regime where boundary-detection paradigms fail, and generalises across organs and modalities by design.

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